HCTZ PBPK/PD model: fit PD

1 Methods: HCTZ PBPK/PD model: fit PD

1.1 Parameters and data

5 parameters were optimized: `E50_hctz_nacl`, `Emax_hctz_na`, `Emax_hctz_cl`, `k_na` and `k_cl`. The parameters of the adopted fit PK_20261017_025702__4167a were fixed at their fitted values.

A fit mapping is a curve of a study, the data and the observable of the model it is compared with. The fit has 77 fit mappings of training data with 166 data points, whose residuals enter the objective; and 8 fit mappings of validation data with 21 data points, which were simulated and evaluated with the fitted parameters but not fitted. The fit was set up and run with sbmlsim [1].

1.2 Objective function

The parameters were estimated by minimizing the objective function

$$F(\vec{p}) = \frac{1}{2} \sum_k \sum_i \rho\left(w_{i,k} r_{i,k}(\vec{p})^2\right) \tag{1}$$

where k runs over the fit mappings of the training data and i over their data points, $r_{i,k}$ is the residual, $w_{i,k} = w_k w^p_{i,k}$ the weight of the data point and ρ the loss function, applied to the squared weighted residual $z = w_{i,k} r^2_{i,k}$. The loss function is linear, $\rho(z) = z$, i.e. a standard least squares problem.

The residuals are normalized by the mean $\overline{y}_k$ of the data of their curve, which makes curves of different quantities and magnitudes comparable:

$$r_{i,k} = \frac{y_{i,k} - f_{k(x_{i,k},\vec{p})}}{\overline{y}_k} \tag{2}$$

The weight of curve k is $w_k = \frac{n_k}{N_k}$, the weight n_k of its fit mapping (the weight its fit mapping collection assigns, 1 by default, or, for a collection which uses the weights of its mappings, the weight of the mapping, by default the number of subjects of its data) divided by the number N_k of its data points.

A data point is weighted with the inverse of its coefficient of variation, $w^p_{i,k} = |y_{i,k} \frac{1}{\sigma_{i,k}}$, with $\sigma_{i,k}$ the standard deviation of the data point, or its standard error for data without a standard deviation. An error of zero is missing, a missing error of a curve with errors is replaced by the largest error of the curve, and the data points of a curve without errors have the weight $w^p_{i,k} = 2$, the weight of a coefficient of variation of 0.5.

1.3 Optimization

The optimizer searched the parameters in log10 space within their bounds; bounds, values and intervals are given in the units of the fit parameters, the units of the table of the parameters.

The objective was minimized in 3 runs with differential evolution [2] of SciPy [3] within the bounds of the parameters, with the strategy `best1bin`, a population of 15 times the number of parameters, at most 1000 generations, the relative tolerance 0.01 of convergence, the recombination constant 0.7 and a mutation dithered between 0.5 and 1. The initial population of a run was drawn by latin hypercube sampling [4], with a seed of every run derived from the seed 1234 of the fit. The best member of the final population was polished with the L-BFGS-B method of SciPy.

The convergence of the runs was assessed from the traces of their cost and the waterfall plot of their final costs, and the parameter set with the lowest cost was used.

The models were simulated with libRoadRunner [5] and its integrator CVODE [6] with the relative tolerance 10^{-6} and the absolute tolerance 10^{-6}, which for a species in a concentration is multiplied by the initial volume of its compartment. It ran with the variable step size of libRoadRunner, i.e. with an output at every step of the integrator. A data point at the time of a change after the start of a simulation, e.g. a dose or the reset of a collection, was compared with the state before the change, except the first data point of a curve with data at later times, which starts a collection and was compared with the state after the change.

1.4 Goodness of fit

The goodness of fit is given per kind of data (training data and validation data) by metrics over its n data points with the residuals $e_i = y_i - f\left(x_i, \hat{\vec{p}}\right)$ at the fitted parameters $\hat{\vec{p}}$, the mean squared error, its root and the root mean squared error normalized by the mean $\overline{y}_{k(i)}$ of the data of the curve of a point:

$$\mathrm{MSE} = \frac{1}{n}\sum_{i=1}^{n} e_i^2, \quad \mathrm{RMSE} = \sqrt{\mathrm{MSE}}, \quad \mathrm{NRMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(\frac{e_i}{\overline{y}_{k(i)}}\right)^2} \tag{3}$$

The root mean square of the residuals of the objective, i.e. what the fit minimizes, and the coefficient of determination:

$$\mathrm{RMSE}_w = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\rho(w_i r_i^2)}, \quad R^2 = 1 - \frac{\sum_i e_i^2}{\sum_i \left(y_i - \overline{y}\right)^2} \tag{4}$$

The predictions of a non-linear model are no linear regression of the data, so R^2 can be negative: the prediction is worse than the mean of the data. The information criteria of Akaike [7] and Schwarz [8] with the number n_p of fitted parameters are those of a least squares fit with normally distributed residuals, up to an additive constant, so only differences between models fitted to the same data are meaningful:

$$\mathrm{AIC} = n\ln(\mathrm{MSE}) + 2n_p, \quad \mathrm{BIC} = n\ln(\mathrm{MSE}) + n_p\ln(n) \tag{5}$$

The percentage of the data points whose prediction $f_i = f\left(x_i, \hat{\vec{p}}\right)$ lies within twofold of the data is taken over the n_+ data points with positive data $y_i > 0$ and a positive prediction $f_i > 0$; the others have no ratio and are not counted, and a kind of data without such a point has no percentage:

$$P_2 = \frac{100}{n_+}\left|\left\{i : \frac{1}{2} \le \frac{f_i}{y_i} \le 2\right\}\right| \tag{6}$$

The goodness of fit plot shows the predictions against the data on logarithmic axes, with the lines of twofold $f = 2y$ and $f = \frac{y}{2}$. The Bland-Altman plot [9] shows the ratio of prediction and data as $d_i = \log_{10}\left(\frac{f_i}{y_i}\right)$ against the geometric mean $\sqrt{f_i y_i}$ for the data points with positive values, with the bias $\overline{d}$ and the limits of agreement $\overline{d} \pm 1.96 s_d$, with the sample standard deviation s_d of d_i, of the training data; they are read as the fold factors $10^{\overline{d}}$ and $10^{\overline{d} \pm 1.96 s_d}$.

1.5 Identifiability

The identifiability of the parameters was analysed with the profile likelihood [10], [11]. The profile of a parameter is the lowest objective with the parameter fixed:

$$\mathrm{PL}(p_j) = \min_{p_{l \ne j}} F(\vec{p}) \tag{7}$$

At every point of a scan the other parameters were re-optimized from the previous point with the least squares optimizer of SciPy with the relative step 0.05. A scan ran from the optimum in both directions in the space of the optimizer, with steps in decades of the parameter, starting at 0.1. A step which raised the objective by more than 0.2 of the distance between the objective at the optimum and the threshold was reduced by the factor 2 and repeated, down to 0.01, and the step after one which raised it by less than that divided by the factor was enlarged by the factor, up to 1. A scan stopped at the threshold, at the bound of the parameter, at a point which failed or after 50 points.

The confidence intervals are the values at which the profile crosses the threshold of the likelihood ratio test

$$F_{\mathrm{thr}} = F_{\mathrm{min}} + \chi^2_\alpha \frac{\mathrm{df}}{2} \tag{8}$$

with F_{min} the lowest objective of the fit and of all profiles and $\chi^2_\alpha(\mathrm{df})$ the quantile of the χ^2 distribution at the confidence level $\alpha = 0.95$ with df = 1 degrees of freedom, i.e. 1.92 above the lowest objective. The threshold has this confidence level if F is the negative log-likelihood of the data up to a constant, i.e. if the weighted residuals are independent and normally distributed with unit variance; otherwise the level is approximate. A crossing is interpolated between two points of the profile, on the logarithm of a parameter on a logarithmic scale. A parameter is identifiable if its profile reaches the threshold within the scan on both sides of the optimum, i.e. its confidence interval is finite; practically non-identifiable towards the lower bound if its profile reaches the threshold above the optimum but not below it within the scan, which ends at the lower bound or after 50 points; practically non-

identifiable towards the upper bound if its profile reaches the threshold below the optimum but not above it within the scan, which ends at the upper bound or after 50 points; practically non-identifiable if its profile reaches the threshold on neither side within the scans, which end at the bounds or after 50 points, and is not flat; and structurally non-identifiable if its profile is flat on both sides. A side of a profile is flat if it rises by less than 0.05 of the distance between the lowest objective and the threshold, and a profile with a point which failed or did not converge is not classified. The profile likelihood is preferred over the intervals of the Fisher information, its intervals are invariant under a transformation of the parameters and can be asymmetric [12].

The Fisher information is the local alternative to the profiles: it is the Gauss-Newton approximation of the curvature of the objective at the optimum, from the Jacobian J of the weighted residuals of the training data with respect to the parameters $\vec{p}$ in the space of the optimizer, computed by central finite differences with the step $h_j = \varepsilon \max(|p_j|, 1),\ \varepsilon = 10^{-6}$ of the parameter p_j:

$$I = J^{\top} J, \quad \mathrm{cov} = \sigma^2 I^{-1}, \quad \sigma^2 = \frac{2F}{n - n_p} \tag{9}$$

with the number n of data points of the training data and the number n_p of parameters. The information has rank 5 of 5, the number of its eigenvalues above 10^{-8} of the largest one, and the condition number 85.5; the covariance of an information of a lower rank is computed with the pseudo-inverse. The Wald intervals are $\hat{p}_j \pm t_{\frac{1+\alpha}{2}, n-n_p} \mathrm{SE}_j$ with the standard errors $\mathrm{SE}_j = \sqrt{\mathrm{cov}_{jj}}$ at the confidence level $\alpha = 0.95$, transformed back into the units of the fit parameters.

1.6 Provenance

The fit PD_20261017_030352__10c1d was run on 2026-10-17 01:03:57 UTC with sbmlsim 0.10.0 [1], libRoadRunner 2.10.0 [5], SciPy 1.18.1 [3], NumPy 2.5.3 [13] and Python 3.14.8 (seed 1234). The models and data were those of commit `02b4b5c535fd42d63a6e025f407656229ea2a747` of `/home/mkoenig/git/pkdb_models` with uncommitted changes; their SHA-256 checksums are listed in Table 1. The fit kept the values `ftissue_hctz` = 0.0637161, `Kp_hctz` = 2.36426, `KI__HCTZEX_f` = 3.23773, `Ka_dis_hctz` = 1.30141 and `GU__HCTZABS_k` = 0.00507166 of the model fixed.

Table 1: SHA-256 checksums of the models and data of the fit PD_20261017_030352__10c1d, their first 12 characters; the full checksums are in the provenance of the result (optimization_result.json).

File	SHA-256
pkdb_models/models/hctz/data/aliskiren/Vaidyanathan2006/.Vaidyanathan2006_Fig2.tsv	4b882f62be16...
pkdb_models/models/hctz/data/aliskiren/Vaidyanathan2006/.Vaidyanathan2006_Fig3.tsv	b5395aea7661...
pkdb_models/models/hctz/data/aliskiren/Vaidyanathan2006/.Vaidyanathan2006_Fig4.tsv	a666a94594f2...
pkdb_models/models/hctz/data/aliskiren/Vaidyanathan2006/.Vaidyanathan2006_Tab2.tsv	e5b023387a7c...
pkdb_models/models/hctz/data/aliskiren/Vaidyanathan2006/.Vaidyanathan2006_Tab3.tsv	6415639e32b1...
pkdb_models/models/hctz/data/aliskiren/Vaidyanathan2006/.Vaidyanathan2006_TabGroups.tsv	e17dd76d2768...
pkdb_models/models/hctz/data/aliskiren/Vaidyanathan2006/.Vaidyanathan2006_TabInterventions.tsv	084628fe174a...
pkdb_models/models/hctz/data/canagliflozin/Devineni2014/.Devineni2014_Fig2.tsv	7ed32853872f...
pkdb_models/models/hctz/data/canagliflozin/Devineni2014/.Devineni2014_Fig3.tsv	ba3c236103c1...
pkdb_models/models/hctz/data/canagliflozin/Devineni2014/.Devineni2014_Tab1.tsv	096cbdc6db9e...
pkdb_models/models/hctz/data/canagliflozin/Devineni2014/.Devineni2014_Tab2.tsv	c5d09da767d6...
pkdb_models/models/hctz/data/canagliflozin/Devineni2014/.Devineni2014_Tab2A.tsv	c6dda93b2bbf...
pkdb_models/models/hctz/data/canagliflozin/Devineni2014/.Devineni2014_TabGroups.tsv	7b29e1f269b4...
pkdb_models/models/hctz/data/canagliflozin/Devineni2014/.Devineni2014_TabInterventions.tsv	e0063bf738d2...
pkdb_models/models/hctz/data/enalapril/Niopas2011/.Niopas2011_Fig1.tsv	a42bd89511f5...

File	SHA-256
pkdb_models/models/hctz/data/enalapril/Niopas2011/.Niopas2011_Fig2.tsv	d28da761910c...
pkdb_models/models/hctz/data/enalapril/Niopas2011/.Niopas2011_Tab2.tsv	083e057dd8d2...
pkdb_models/models/hctz/data/enalapril/Niopas2011/.Niopas2011_Tab3.tsv	8b7e9d21a561...
pkdb_models/models/hctz/data/enalapril/Niopas2011/.Niopas2011_TabGroups.tsv	0264a6f8c5c8...
pkdb_models/models/hctz/data/enalapril/Niopas2011/.Niopas2011_TabInterventions.tsv	7e44753b6aa6...
pkdb_models/models/hctz/data/hydrochlorothiazide/Anderson1961/.Anderson1961_Tab1.tsv	0b96662f4794...
pkdb_models/models/hctz/data/hydrochlorothiazide/Anderson1961/.Anderson1961_Tab1A.tsv	ddb7c92add6e...
pkdb_models/models/hctz/data/hydrochlorothiazide/Anderson1961/.Anderson1961_Tab1B.tsv	5dad76e537a1...
pkdb_models/models/hctz/data/hydrochlorothiazide/Anderson1961/.Anderson1961_TabGroups.tsv	1d517cac3d89...
pkdb_models/models/hctz/data/hydrochlorothiazide/Anderson1961/.Anderson1961_TabIndividuals.tsv	9468bc219f5e...
pkdb_models/models/hctz/data/hydrochlorothiazide/Anderson1961/.Anderson1961_TabInterventions.tsv	c3245d431e24...
pkdb_models/models/hctz/data/hydrochlorothiazide/Azumaya1990/.Azumaya1990_Fig3.tsv	ead505237a28...
pkdb_models/models/hctz/data/hydrochlorothiazide/Azumaya1990/.Azumaya1990_TabGroups.tsv	3d24752729ca...
pkdb_models/models/hctz/data/hydrochlorothiazide/Azumaya1990/.Azumaya1990_TabInterventions.tsv	9d6d47aeb522...
pkdb_models/models/hctz/data/hydrochlorothiazide/Barbhaiya1982/.Barbhaiya1982_Fig1.tsv	4f85c3948a47...
pkdb_models/models/hctz/data/hydrochlorothiazide/Barbhaiya1982/.Barbhaiya1982_Fig2.tsv	1a0c037cf194...
pkdb_models/models/hctz/data/hydrochlorothiazide/Barbhaiya1982/.Barbhaiya1982_Fig3.tsv	c5480ff6b536...
pkdb_models/models/hctz/data/hydrochlorothiazide/Barbhaiya1982/.Barbhaiya1982_Fig5.tsv	2367b86e69cb...
pkdb_models/models/hctz/data/hydrochlorothiazide/Barbhaiya1982/.Barbhaiya1982_Tab1.tsv	71d41f5bf888...
pkdb_models/models/hctz/data/hydrochlorothiazide/Barbhaiya1982/.Barbhaiya1982_TabGroups.tsv	1c678a4589a5...
pkdb_models/models/hctz/data/hydrochlorothiazide/Barbhaiya1982/.Barbhaiya1982_TabInterventions.tsv	069e892ea59b...
pkdb_models/models/hctz/data/hydrochlorothiazide/Barbhaiya1982a/.Barbhaiya1982a_Tab1.tsv	e910986f2e22...
pkdb_models/models/hctz/data/hydrochlorothiazide/Barbhaiya1982a/.Barbhaiya1982a_Tab2.tsv	3dbbeef29ba0...
pkdb_models/models/hctz/data/hydrochlorothiazide/Barbhaiya1982a/.Barbhaiya1982a_Tab3.tsv	e3183d391872...
pkdb_models/models/hctz/data/hydrochlorothiazide/Barbhaiya1982a/.Barbhaiya1982a_TabGroups.tsv	c227d8c7b0f8...
pkdb_models/models/hctz/data/hydrochlorothiazide/Barbhaiya1982a/.Barbhaiya1982a_TabInterventions.tsv	3c6fd578a468...
pkdb_models/models/hctz/data/hydrochlorothiazide/Beermann1976/.Beermann1976_Fig2.tsv	2807904243da...

File	SHA-256
pkdb_models/models/hctz/data/hydrochlorothiazide/Beermann1976/.Beermann1976_Fig3.tsv	d5ce6e1099f8...
pkdb_models/models/hctz/data/hydrochlorothiazide/Beermann1976/.Beermann1976_Tab1.tsv	fb16ac2a9245...
pkdb_models/models/hctz/data/hydrochlorothiazide/Beermann1976/.Beermann1976_Tab1A.tsv	d5a0f6178115...
pkdb_models/models/hctz/data/hydrochlorothiazide/Beermann1976/.Beermann1976_TabGroups.tsv	e1b9eecde8ab...
pkdb_models/models/hctz/data/hydrochlorothiazide/Beermann1976/.Beermann1976_TabIndividuals.tsv	7bdd5f3efc78...
pkdb_models/models/hctz/data/hydrochlorothiazide/Beermann1976/.Beermann1976_TabInterventions.tsv	756ad10cf430...
pkdb_models/models/hctz/data/hydrochlorothiazide/Beermann1977a/.Beermann1977a_Fig1_2.tsv	8135a1aef6a4...
pkdb_models/models/hctz/data/hydrochlorothiazide/Beermann1977a/.Beermann1977a_Fig3.tsv	9b8aecc3043d...
pkdb_models/models/hctz/data/hydrochlorothiazide/Beermann1977a/.Beermann1977a_Fig4.tsv	45eb63754276...
pkdb_models/models/hctz/data/hydrochlorothiazide/Beermann1977a/.Beermann1977a_Fig5.tsv	ce4223f8d147...
pkdb_models/models/hctz/data/hydrochlorothiazide/Beermann1977a/.Beermann1977a_Fig5tc.tsv	ef1a309635e7...
pkdb_models/models/hctz/data/hydrochlorothiazide/Beermann1977a/.Beermann1977a_Tab3.tsv	9ed1f48b705a...
pkdb_models/models/hctz/data/hydrochlorothiazide/Beermann1977a/.Beermann1977a_Tab4.tsv	64c35907cf12...
pkdb_models/models/hctz/data/hydrochlorothiazide/Beermann1977a/.Beermann1977a_Tab4mean.tsv	42a9478560e8...
pkdb_models/models/hctz/data/hydrochlorothiazide/Beermann1977a/.Beermann1977a_TabGroups.tsv	e1a2a61ff2cd...
pkdb_models/models/hctz/data/hydrochlorothiazide/Beermann1977a/.Beermann1977a_TabIndividuals.tsv	d02c16eca30a...
pkdb_models/models/hctz/data/hydrochlorothiazide/Beermann1977a/.Beermann1977a_TabInterventions.tsv	0dbb93f4c798...
pkdb_models/models/hctz/data/hydrochlorothiazide/Beermann1979/.Beermann1979_Fig1.tsv	77696706cdc8...
pkdb_models/models/hctz/data/hydrochlorothiazide/Beermann1979/.Beermann1979_Tab2.tsv	00ecc7bc09f2...
pkdb_models/models/hctz/data/hydrochlorothiazide/Beermann1979/.Beermann1979_Tab3.tsv	f08b3337f300...
pkdb_models/models/hctz/data/hydrochlorothiazide/Beermann1979/.Beermann1979_TabGroups.tsv	eb6901a5ec90...
pkdb_models/models/hctz/data/hydrochlorothiazide/Beermann1979/.Beermann1979_TabIndividuals.tsv	eae34c5a0b0e...
pkdb_models/models/hctz/data/hydrochlorothiazide/Beermann1979/.Beermann1979_TabInterventions.tsv	05987a4ccfb5...
pkdb_models/models/hctz/data/hydrochlorothiazide/Dussol2005/.Dussol2005_Tab2.tsv	75b8326b1131...
pkdb_models/models/hctz/data/hydrochlorothiazide/Dussol2005/.Dussol2005_Tab3.tsv	cd2268796116...
pkdb_models/models/hctz/data/hydrochlorothiazide/Dussol2005/.Dussol2005_TabGroups.tsv	68166df9b390...

File	SHA-256
pkdb_models/models/hctz/data/hydrochlorothiazide/Dussol2005/.Dussol2005_TabIndividuals.tsv	b7872677dd09...
pkdb_models/models/hctz/data/hydrochlorothiazide/Dussol2005/.Dussol2005_TabInterventions.tsv	02019f8d2382...
pkdb_models/models/hctz/data/hydrochlorothiazide/Ernst2006/.Ernst2006_Fig1.tsv	a994fecc105d...
pkdb_models/models/hctz/data/hydrochlorothiazide/Giudicelli1987/.Giudicelli1987_Fig1.tsv	a7441ff279b1...
pkdb_models/models/hctz/data/hydrochlorothiazide/Giudicelli1987/.Giudicelli1987_Fig2.tsv	eb43d99d1abf...
pkdb_models/models/hctz/data/hydrochlorothiazide/Giudicelli1987/.Giudicelli1987_Fig3.tsv	18912daaff30...
pkdb_models/models/hctz/data/hydrochlorothiazide/Giudicelli1987/.Giudicelli1987_Fig4.tsv	ff67b5f49cbc...
pkdb_models/models/hctz/data/hydrochlorothiazide/Giudicelli1987/.Giudicelli1987_Fig5.tsv	ee2adf35ff0b...
pkdb_models/models/hctz/data/hydrochlorothiazide/Giudicelli1987/.Giudicelli1987_Tab2.tsv	06b166899abe...
pkdb_models/models/hctz/data/hydrochlorothiazide/Giudicelli1987/.Giudicelli1987_Tab4.tsv	4b470bc80366...
pkdb_models/models/hctz/data/hydrochlorothiazide/Giudicelli1987/.Giudicelli1987_TabGroups.tsv	29d958be9670...
pkdb_models/models/hctz/data/hydrochlorothiazide/Giudicelli1987/.Giudicelli1987_TabIndividuals.tsv	ef74d1ac3e12...
pkdb_models/models/hctz/data/hydrochlorothiazide/Giudicelli1987/.Giudicelli1987_TabInterventions.tsv	9582c22f6643...
pkdb_models/models/hctz/data/hydrochlorothiazide/Heise2015/.Heise2015_Fig2.tsv	f4935dc54371...
pkdb_models/models/hctz/data/hydrochlorothiazide/Heise2015/.Heise2015_Fig3.tsv	3f208e2a31a5...
pkdb_models/models/hctz/data/hydrochlorothiazide/Heise2015/.Heise2015_Fig4.tsv	8a0cb6a070e5...
pkdb_models/models/hctz/data/hydrochlorothiazide/Heise2015/.Heise2015_Tab1.tsv	f1f21b9bf441...
pkdb_models/models/hctz/data/hydrochlorothiazide/Heise2015/.Heise2015_Tab3.tsv	b87e4c93ab81...
pkdb_models/models/hctz/data/hydrochlorothiazide/Heise2015/.Heise2015_Tab3A.tsv	0de829681d4f...
pkdb_models/models/hctz/data/hydrochlorothiazide/Heise2015/.Heise2015_Tab4.tsv	1f6fecb5eea0...
pkdb_models/models/hctz/data/hydrochlorothiazide/Heise2015/.Heise2015_TabGroups.tsv	d77f4251d594...
pkdb_models/models/hctz/data/hydrochlorothiazide/Heise2015/.Heise2015_TabInterventions.tsv	36a7fdc8e888...
pkdb_models/models/hctz/data/hydrochlorothiazide/Howes1991/.Howes1991_Fig1.tsv	87a1bcadd50e...
pkdb_models/models/hctz/data/hydrochlorothiazide/Howes1991/.Howes1991_Tab1.tsv	8a9ea4517461...
pkdb_models/models/hctz/data/hydrochlorothiazide/Howes1991/.Howes1991_Tab2.tsv	82f669940103...
pkdb_models/models/hctz/data/hydrochlorothiazide/Howes1991/.Howes1991_Tab3.tsv	5b8962c88ee3...
pkdb_models/models/hctz/data/hydrochlorothiazide/Howes1991/.Howes1991_TabGroups.tsv	67c05dd8e03d...
pkdb_models/models/hctz/data/hydrochlorothiazide/Howes1991/.Howes1991_TabInterventions.tsv	02941e5da0c2...
pkdb_models/models/hctz/data/hydrochlorothiazide/Hsiao2015/.Hsiao2015_Fig1.tsv	64fd8ab77b40...
pkdb_models/models/hctz/data/hydrochlorothiazide/Hsiao2015/.Hsiao2015_Tab1.tsv	00ce0500feca...
pkdb_models/models/hctz/data/hydrochlorothiazide/Hsiao2015/.Hsiao2015_TabGroups.tsv	564ae62fbfd3...

File	SHA-256
pkdb_models/models/hctz/data/hydrochlorothiazide/Hsiao2015/.Hsiao2015_TabInterventions.tsv	a5793f71d672...
pkdb_models/models/hctz/data/hydrochlorothiazide/Hunninghake1986/.Hunninghake1986_Fig1.tsv	b784bcb3e9ad...
pkdb_models/models/hctz/data/hydrochlorothiazide/Hunninghake1986/.Hunninghake1986_Fig2.tsv	c76172d843b0...
pkdb_models/models/hctz/data/hydrochlorothiazide/Hunninghake1986/.Hunninghake1986_Fig3.tsv	394c3b42e8e4...
pkdb_models/models/hctz/data/hydrochlorothiazide/Hunninghake1986/.Hunninghake1986_Tab2.tsv	0f92641bead3...
pkdb_models/models/hctz/data/hydrochlorothiazide/Hunninghake1986/.Hunninghake1986_TabGroups.tsv	23883b9fb900...
pkdb_models/models/hctz/data/hydrochlorothiazide/Hunninghake1986/.Hunninghake1986_TabInterventions.tsv	68dde1a2a141...
pkdb_models/models/hctz/data/hydrochlorothiazide/Januszewicz1959/.Januszewicz1959_Fig2tc.tsv	44fc27e9b75f...
pkdb_models/models/hctz/data/hydrochlorothiazide/Januszewicz1959/.Januszewicz1959_Tab1.tsv	df66df0654ff...
pkdb_models/models/hctz/data/hydrochlorothiazide/Januszewicz1959/.Januszewicz1959_TabGroups.tsv	b432c9f59ea3...
pkdb_models/models/hctz/data/hydrochlorothiazide/Januszewicz1959/.Januszewicz1959_TabIndividuals.tsv	c6d9b60bbaa6...
pkdb_models/models/hctz/data/hydrochlorothiazide/Januszewicz1959/.Januszewicz1959_TabInterventions.tsv	fcb6d290080b...
pkdb_models/models/hctz/data/hydrochlorothiazide/Jeon2012/.Jeon2012_Fig3.tsv	5d13f75a5a76...
pkdb_models/models/hctz/data/hydrochlorothiazide/Jeon2012/.Jeon2012_Fig4.tsv	8041d6538ced...
pkdb_models/models/hctz/data/hydrochlorothiazide/Jeon2012/.Jeon2012_Fig5.tsv	c98f449945e9...
pkdb_models/models/hctz/data/hydrochlorothiazide/Jeon2012/.Jeon2012_Fig6.tsv	77ead1fbfcdd...
pkdb_models/models/hctz/data/hydrochlorothiazide/Jeon2012/.Jeon2012_Tab2.tsv	4cb2bde69cc3...
pkdb_models/models/hctz/data/hydrochlorothiazide/Jeon2012/.Jeon2012_TabGroups.tsv	4ed151ca9458...
pkdb_models/models/hctz/data/hydrochlorothiazide/Jeon2012/.Jeon2012_TabInterventions.tsv	2743ddcea85b...
pkdb_models/models/hctz/data/hydrochlorothiazide/Jordo1979/.Jordo1979_Fig1.tsv	eecd7245850d...
pkdb_models/models/hctz/data/hydrochlorothiazide/Jordo1979/.Jordo1979_Fig2.tsv	f41b81f187f2...
pkdb_models/models/hctz/data/hydrochlorothiazide/Jordo1979/.Jordo1979_Tab1.tsv	cddc1c86b8b9...
pkdb_models/models/hctz/data/hydrochlorothiazide/Jordo1979/.Jordo1979_Tab2.tsv	0379d6aaaa5a...
pkdb_models/models/hctz/data/hydrochlorothiazide/Jordo1979/.Jordo1979_Tab2Mean.tsv	8d876091a4eb...
pkdb_models/models/hctz/data/hydrochlorothiazide/Jordo1979/.Jordo1979_TabGroups.tsv	9daa6d6555b0...
pkdb_models/models/hctz/data/hydrochlorothiazide/Jordo1979/.Jordo1979_TabIndividuals.tsv	679da5cce1b0...
pkdb_models/models/hctz/data/hydrochlorothiazide/Jordo1979/.Jordo1979_TabInterventions.tsv	06d6eca836c2...
pkdb_models/models/hctz/data/hydrochlorothiazide/Knauf1995/.Knauf1995_Fig1.tsv	fbcd718e5f2c...
pkdb_models/models/hctz/data/hydrochlorothiazide/Knauf1995/.Knauf1995_Fig2.tsv	dc5617d97660...
pkdb_models/models/hctz/data/hydrochlorothiazide/Knauf1995/.Knauf1995_Fig3.tsv	5763164aec01...
pkdb_models/models/hctz/data/hydrochlorothiazide/Knauf1995/.Knauf1995_Fig4.tsv	911aa1a1e89d...

File	SHA-256
pkdb_models/models/hctz/data/hydrochlorothiazide/Knauf1995/.Knauf1995_Fig5.tsv	6141428da0e1...
pkdb_models/models/hctz/data/hydrochlorothiazide/Knauf1995/.Knauf1995_Tab2.tsv	a62e96697a27...
pkdb_models/models/hctz/data/hydrochlorothiazide/Knauf1995/.Knauf1995_TabGroups.tsv	cec858380cf8...
pkdb_models/models/hctz/data/hydrochlorothiazide/Knauf1995/.Knauf1995_TabIndividuals.tsv	7311c65ddbec...
pkdb_models/models/hctz/data/hydrochlorothiazide/Knauf1995/.Knauf1995_TabInterventions.tsv	65762172be82...
pkdb_models/models/hctz/data/hydrochlorothiazide/Niemeyer1983/.Niemeyer1983_Fig1.tsv	58c88431c8b8...
pkdb_models/models/hctz/data/hydrochlorothiazide/Niemeyer1983/.Niemeyer1983_Fig2.tsv	f730db99af80...
pkdb_models/models/hctz/data/hydrochlorothiazide/Niemeyer1983/.Niemeyer1983_Fig3.tsv	fe160c3a04cc...
pkdb_models/models/hctz/data/hydrochlorothiazide/Niemeyer1983/.Niemeyer1983_Tab3.tsv	36768c2c955c...
pkdb_models/models/hctz/data/hydrochlorothiazide/Niemeyer1983/.Niemeyer1983_TabGroups.tsv	17d5aa23ed83...
pkdb_models/models/hctz/data/hydrochlorothiazide/Niemeyer1983/.Niemeyer1983_TabIndividuals.tsv	9d95d6859878...
pkdb_models/models/hctz/data/hydrochlorothiazide/Niemeyer1983/.Niemeyer1983_TabInterventions.tsv	a9993f8762dd...
pkdb_models/models/hctz/data/hydrochlorothiazide/Nilsen1989/.Nilsen1989_Fig1.tsv	f9095f95471e...
pkdb_models/models/hctz/data/hydrochlorothiazide/Nilsen1989/.Nilsen1989_Fig2.tsv	71223ba3b02f...
pkdb_models/models/hctz/data/hydrochlorothiazide/Nilsen1989/.Nilsen1989_Fig3.tsv	e5c1f8ad9c2b...
pkdb_models/models/hctz/data/hydrochlorothiazide/Nilsen1989/.Nilsen1989_Fig4.tsv	b4be3f4c6da0...
pkdb_models/models/hctz/data/hydrochlorothiazide/Nilsen1989/.Nilsen1989_Fig5.tsv	f982121b79f6...
pkdb_models/models/hctz/data/hydrochlorothiazide/Nilsen1989/.Nilsen1989_Tab1.tsv	db3193becc56...
pkdb_models/models/hctz/data/hydrochlorothiazide/Nilsen1989/.Nilsen1989_TabGroups.tsv	4d9afd8ecc3c...
pkdb_models/models/hctz/data/hydrochlorothiazide/Nilsen1989/.Nilsen1989_TabInterventions.tsv	1ffefd2c38cf...
pkdb_models/models/hctz/data/hydrochlorothiazide/Patel1984/.Patel1984_Fig1.tsv	492138f96021...
pkdb_models/models/hctz/data/hydrochlorothiazide/Patel1984/.Patel1984_Fig2.tsv	146b2c18c2bd...
pkdb_models/models/hctz/data/hydrochlorothiazide/Patel1984/.Patel1984_Tab2.tsv	7007b5bbbe0e...
pkdb_models/models/hctz/data/hydrochlorothiazide/Patel1984/.Patel1984_Tab4.tsv	d6c5d9e1dee9...
pkdb_models/models/hctz/data/hydrochlorothiazide/Patel1984/.Patel1984_TabGroups.tsv	cf0bada6496e...
pkdb_models/models/hctz/data/hydrochlorothiazide/Patel1984/.Patel1984_TabInterventions.tsv	a7448276e496...
pkdb_models/models/hctz/data/hydrochlorothiazide/Ripley2000/.Ripley2000_Fig1_2.tsv	349ae182f694...
pkdb_models/models/hctz/data/hydrochlorothiazide/Ripley2000/.Ripley2000_Fig2.tsv	f944d94feb78...
pkdb_models/models/hctz/data/hydrochlorothiazide/Ripley2000/.Ripley2000_Fig3.tsv	20e3805d7769...
pkdb_models/models/hctz/data/hydrochlorothiazide/Ripley2000/.Ripley2000_Tab3.tsv	e3871f9fabdc...
pkdb_models/models/hctz/data/hydrochlorothiazide/Ripley2000/.Ripley2000_TabGroups.tsv	223f701288a6...
pkdb_models/models/hctz/data/hydrochlorothiazide/Ripley2000/.Ripley2000_TabInterventions.tsv	cf973e21bbcb...

File	SHA-256
pkdb_models/models/hctz/data/hydrochlorothiazide/Weir1998/.Weir1998_Fig2.tsv	4d781066aac0...
pkdb_models/models/hctz/data/hydrochlorothiazide/Weir1998/.Weir1998_Fig3.tsv	c3ba0214b53b...
pkdb_models/models/hctz/data/hydrochlorothiazide/Weir1998/.Weir1998_Tab3.tsv	65c6ff1c8c7a...
pkdb_models/models/hctz/data/hydrochlorothiazide/Weir1998/.Weir1998_Tab4.tsv	822c640eed5a...
pkdb_models/models/hctz/data/hydrochlorothiazide/Weir1998/.Weir1998_TabGroups.tsv	a52f9f3880a7...
pkdb_models/models/hctz/data/hydrochlorothiazide/Weir1998/.Weir1998_TabInterventions.tsv	9c3826e429da...
pkdb_models/models/hctz/data/hydrochlorothiazide/Williams1982/.Williams1982_Fig2.tsv	cfda53e0bc71...
pkdb_models/models/hctz/data/hydrochlorothiazide/Williams1982/.Williams1982_Fig2A.tsv	8d9332b11962...
pkdb_models/models/hctz/data/hydrochlorothiazide/Williams1982/.Williams1982_Fig2B.tsv	2bc1ea3c8faa...
pkdb_models/models/hctz/data/hydrochlorothiazide/Williams1982/.Williams1982_Fig2C.tsv	7f28e773d2dc...
pkdb_models/models/hctz/data/hydrochlorothiazide/Williams1982/.Williams1982_Fig2sc.tsv	8d9332b11962...
pkdb_models/models/hctz/data/hydrochlorothiazide/Williams1982/.Williams1982_Fig3.tsv	b7f69903770e...
pkdb_models/models/hctz/data/hydrochlorothiazide/Williams1982/.Williams1982_Fig3A.tsv	2e4214b536ca...
pkdb_models/models/hctz/data/hydrochlorothiazide/Williams1982/.Williams1982_Fig3B.tsv	c1a6d5b88f7a...
pkdb_models/models/hctz/data/hydrochlorothiazide/Williams1982/.Williams1982_Tab1.tsv	7880d1048cd6...
pkdb_models/models/hctz/data/hydrochlorothiazide/Williams1982/.Williams1982_Tab2.tsv	142ccecd0aac...
pkdb_models/models/hctz/data/hydrochlorothiazide/Williams1982/.Williams1982_TabGroups.tsv	be5b1f862ad7...
pkdb_models/models/hctz/data/hydrochlorothiazide/Williams1982/.Williams1982_TabInterventions.tsv	bd50b21b1457...
pkdb_models/models/hctz/data/lisinopril/Koytchev2004/.Koytchev2004_Fig1.tsv	5ed8ba49e6dc...
pkdb_models/models/hctz/data/lisinopril/Koytchev2004/.Koytchev2004_Fig2.tsv	8d2e06479531...
pkdb_models/models/hctz/data/lisinopril/Koytchev2004/.Koytchev2004_Tab1.tsv	86e50f974b9f...
pkdb_models/models/hctz/data/lisinopril/Koytchev2004/.Koytchev2004_Tab3.tsv	742f83ae1f39...
pkdb_models/models/hctz/data/lisinopril/Koytchev2004/.Koytchev2004_TabGroups.tsv	9a4ff71bdd31...
pkdb_models/models/hctz/data/lisinopril/Koytchev2004/.Koytchev2004_TabInterventions.tsv	1b45614b4423...
pkdb_models/models/hctz/models/output/models/hctz_body_flat.xml	a8c0d649dbe1...

2 Results: HCTZ PBPK/PD model: fit PD

2.1 Parameters

Table 2: Parameters of the fit PD_20261017_030352__10c1d, the parameter set PD_20261017_030352__10c1d_0, cost 65.1, with their bounds in the optimization. The 95% confidence intervals are those of the profile likelihood, LB and UB an interval which is open towards the lower or the upper bound, and the Wald intervals of the Fisher information for a parameter without a profile. Identifiable is the classification of the profile: yes, no (LB) or no (UB) if the profile does not reach the threshold towards the lower or the upper bound, no if it reaches it on neither side, structural if it is flat.

Parameter	Description	Value [95% CI]	Unit	Bounds	Identifiable
E50_hctz_nacl		8.62e-05 [2.86e-05, 0.000289]	mmol/l	[1e-06, 0.01]	yes
Emax_hctz_na		1.73 [1.25, 2.6]	dimensionless	[0.1, 5]	yes
Emax_hctz_cl		0.457 [0.312, 0.768]	dimensionless	[0.1, 5]	yes
k_na		0.000677 [0.000584, 0.000788]	l/min	[0.00025, 0.0015]	yes
k_cl		0.00371 [0.00252, 0.00512]	l/min	[0.00034, 0.007]	yes

2.2 Data

Table 3: Data of the fit PD_20261017_030352__10c1d: the number of fit mappings and, in parentheses, of data points per study and kind. The training data enters the cost, the validation data and the outliers are evaluated only.

Study	Training	Validation	Outlier	Total
Beermann1977a	5 (5)	5 (5)	-	10 (10)
Jeon2012	3 (17)	-	-	3 (17)
Knauf1995	48 (48)	1 (14)	-	49 (62)
Nilsen1989	1 (6)	-	-	1 (6)
Patel1984	10 (10)	2 (2)	-	12 (12)
Ripley2000	4 (8)	-	-	4 (8)
Williams1982	6 (72)	-	-	6 (72)
Total	77 (166)	8 (21)	-	85 (187)

2.3 Goodness of fit

Table 4: Goodness of fit of the fit PD_20261017_030352__10c1d per kind of data, the parameter set PD_20261017_030352__10c1d_0. The cost is defined on the training data only, 2-fold is the percentage of the data points with a positive value and prediction whose prediction is within twofold of the data.

Kind	n	cost	MSE	RMSE	NRMSE	RMSE_w	R^2	2-fold [%]	AIC	BIC
training	166	65.1	996	31.6	0.514	0.886	0.785	88.8	1156.0	1171.5
validation	21	-	2.33e+03	48.3	0.275	1.29	0.875	100.0	172.8	178.0

2.4 Figures

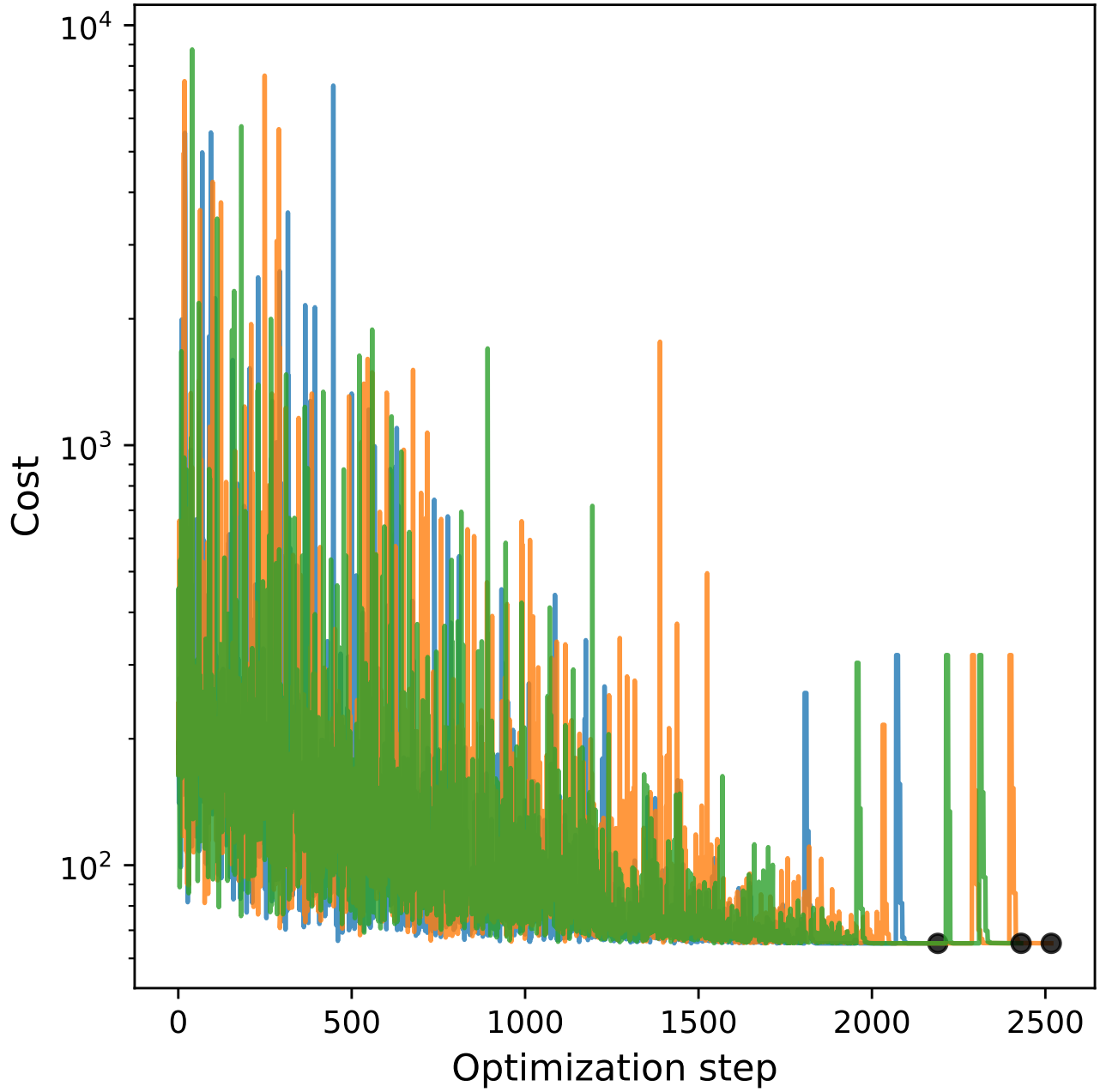


Figure 1: Cost of the optimizers over their steps, fit PD_20261017_030352__10c1d. The cost of every optimization run over its steps. Runs which end at the same cost found the same optimum; a run which stops much higher is stuck in a local minimum.

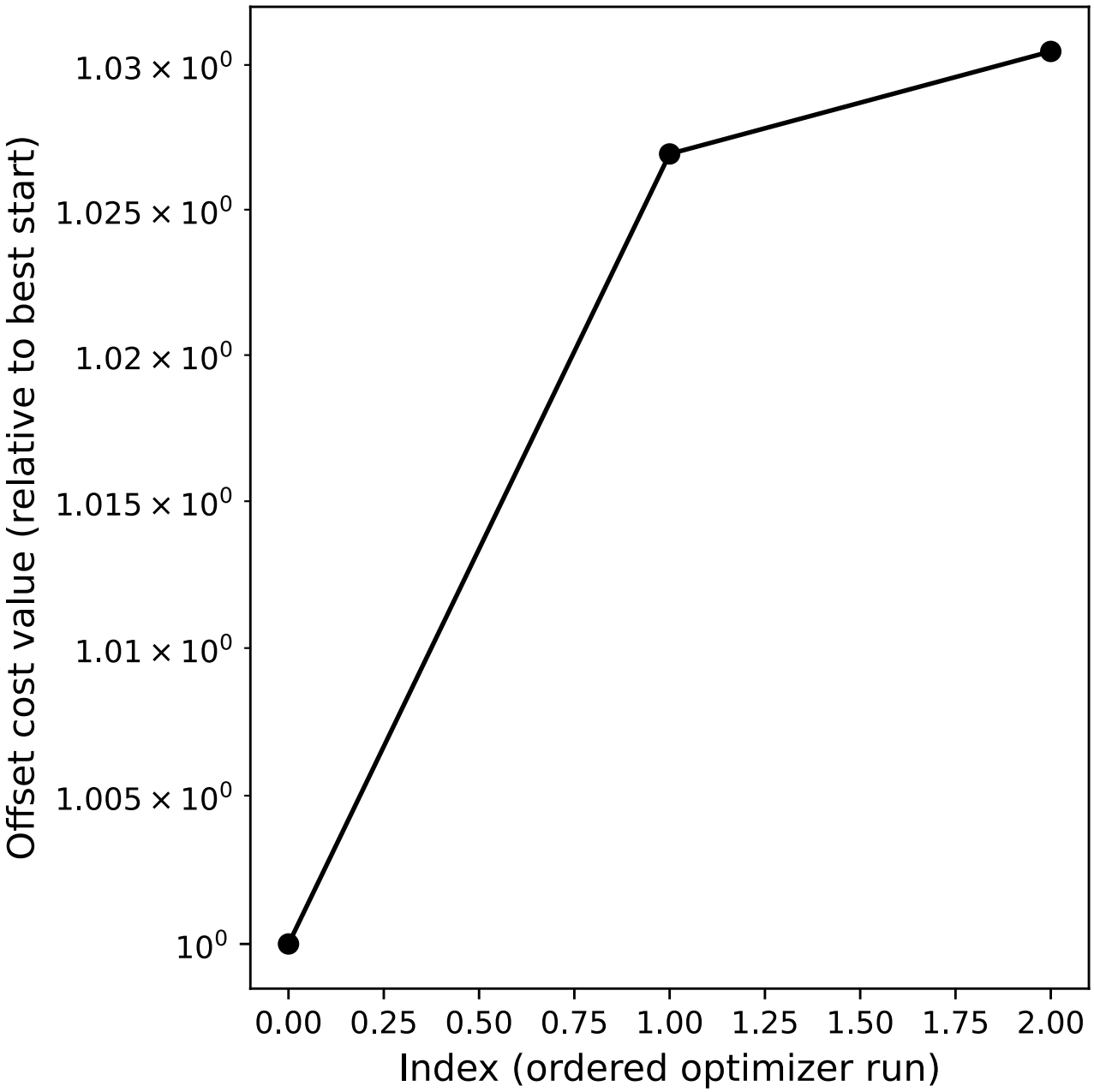


Figure 2: Cost of the optimization runs, ordered, fit PD_20261017_030352__10c1d. The final cost of the runs, ordered. A flat plateau at the left is the global optimum found repeatedly, which is the evidence that the multistart converged; steps to the right are local minima.

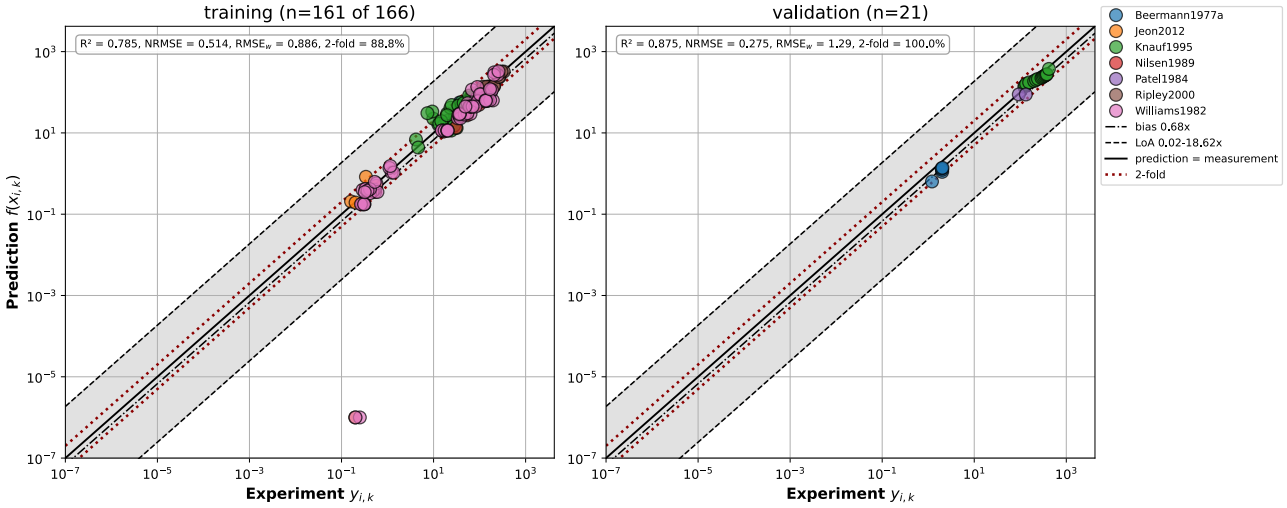


Figure 3: Prediction against the measured data points, per kind of fit mapping, with the agreement of the training data, the lines of twofold and the metrics of the panel, fit PD_20261017_030352__10c1d. Within twofold of the data are 143 of the 161 training data points (88.8%) and 21 of the 21 validation data points (100.0%); only the data points with a positive value and prediction are counted. Prediction against measurement, one point per data point, with a panel per kind of fit mapping. The points scatter around the diagonal when the model describes the data; a systematic deviation from it is a systematic error of the model. The band is the agreement of the training data, i.e. the same band the Bland-Altman plot draws, here as lines parallel to the diagonal. The dark red dotted lines are twofold, prediction = 2 x measurement and prediction = measurement / 2; the caption states the points between them of every kind. A point which is zero or negative is not drawn on the logarithmic axes and not counted, the title of a panel then states the points drawn of all. The validation panel shows how the fit describes data it was not fitted on, the outlier panel the data the fit dropped. The box of a panel carries its R², its normalized RMSE, the RMSE of the residuals of the cost and the percentage of its points within twofold.

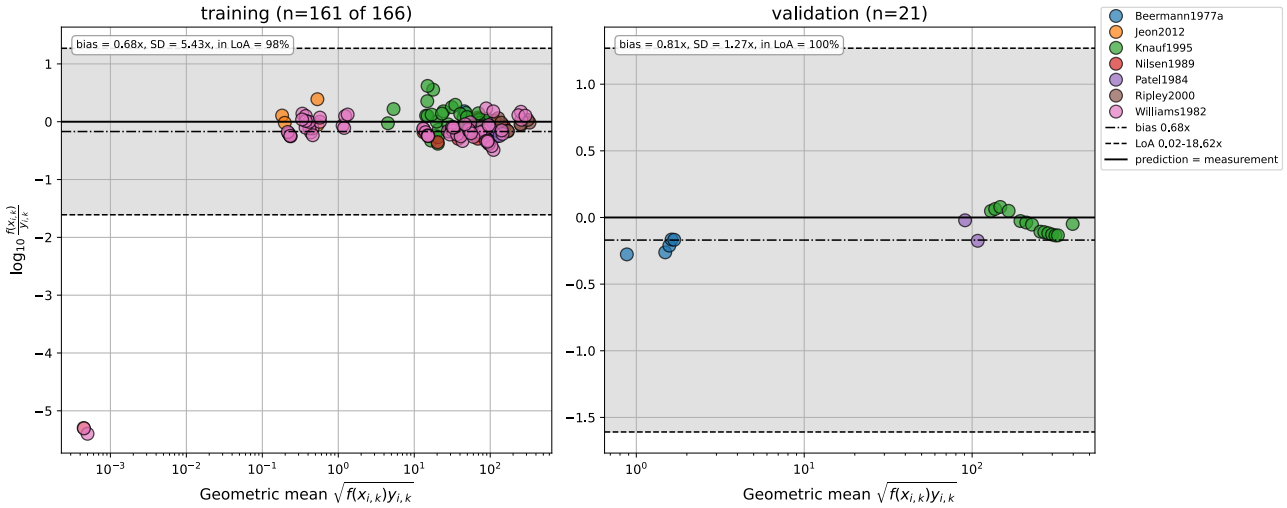


Figure 4: Agreement of prediction and measurement as a ratio, per kind of fit mapping, with the agreement of the training data and the agreement of the panel, fit PD_20261017_030352__10c1d. The ratio of prediction and measurement over the geometric mean of the two, i.e. the goodness of fit with the diagonal turned into the horizontal. The band is the same, the bias and the limits of agreement `bias ± 1.96 SD` of the training data as fold factors. A bias away from 1 is a systematic over- or underprediction, wide limits are a large scatter, and a trend over the mean is a model which describes the large or the small values better. The box of a panel carries the bias and the SD of its own points as fold factors and the share of them inside the limits of agreement of the training data.

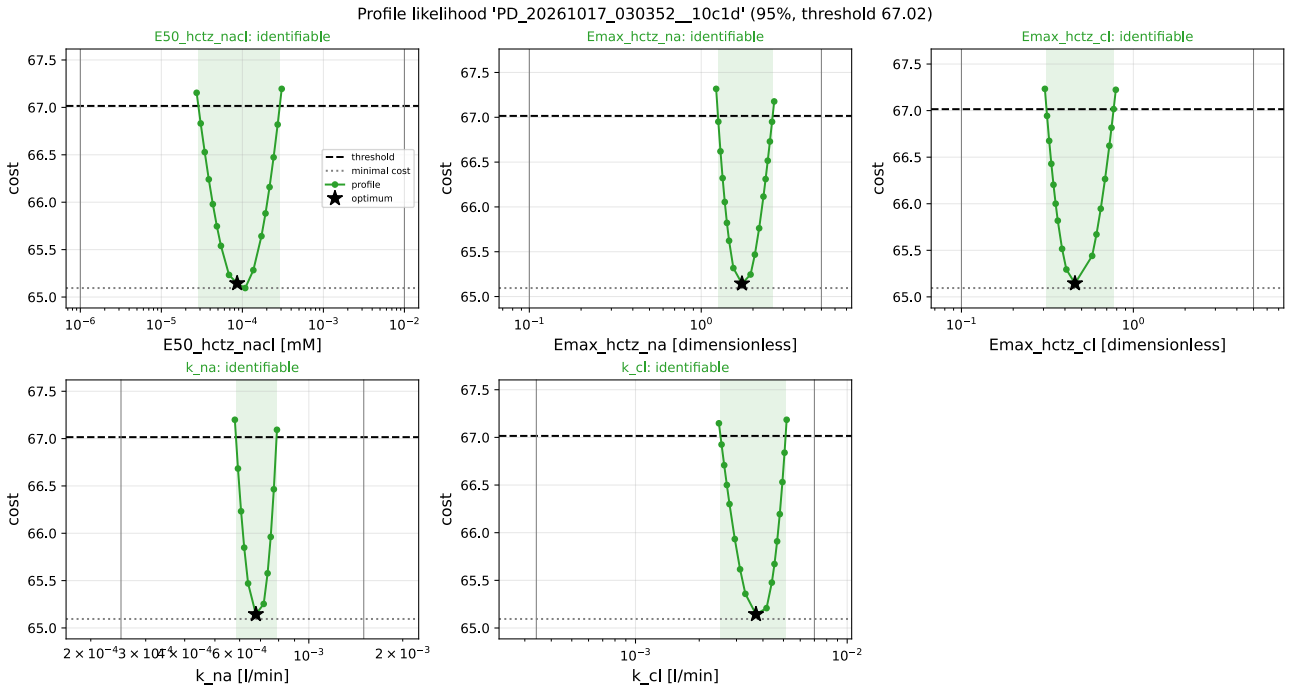


Figure 5: Profile likelihood of every parameter: the cost with the parameter fixed and the other parameters optimized, the threshold of the confidence level and the confidence interval, fit PD_20261017_030352__10c1d. One profile per parameter. A profile which crosses the dashed threshold on both sides gives a finite confidence interval; one which stays below it up to a bound is practically non-identifiable; a flat profile is structurally non-identifiable, the parameter is compensated by the others.

2.5 Provenance

Table 5: Provenance of the fit PD_20261017_030352__10c1d: the versions it ran with and the revision of its models and data; the SHA-256 checksums of the models and data are listed in Table 1.

Item	Value
Fit	`PD_20261017_030352__10c1d`
Created	2026-10-17 01:03:57 UTC
Seed	1234
Python	3.14.8
sbmlsim	0.10.0
libRoadRunner	2.10.0
SciPy	1.18.1
NumPy	2.5.3
Repository	`/home/mkoenig/git/pkdb_models`
Commit	`02b4b5c535fd42d63a6e025f407656229ea2a747`
Uncommitted changes	yes

Bibliography

[1] M. König, “sbmlsim: SBML simulation made easy.” Zenodo, 2026. doi: 10.5281/zenodo.3597770. Version 0.10.0.

[2] R. Storn and K. Price, “Differential evolution - a simple and efficient heuristic for global optimization over continuous spaces,” *Journal of Global Optimization*, vol. 11, pp. 341–359, 1997, doi: 10.1023/A:1008202821328.

[3] P. Virtanen *et al.*, “SciPy 1.0: fundamental algorithms for scientific computing in Python,” *Nature Methods*, vol. 17, pp. 261–272, 2020, doi: 10.1038/s41592-019-0686-2. Version 1.18.1.

[4] M. D. McKay, R. J. Beckman, and W. J. Conover, “A comparison of three methods for selecting values of input variables in the analysis of output from a computer code,” *Technometrics*, vol. 21, no. 2, pp. 239–245, 1979, doi: 10.2307/1268522.

[5] C. Welsh, J. Xu, L. Smith, M. König, K. Choi, and H. M. Sauro, "libRoadRunner 2.0: a high performance SBML simulation and analysis library," *Bioinformatics*, vol. 39, no. 1, p. btac770, 2023, doi: 10.1093/bioinformatics/btac770. Version 2.10.0.

[6] A. C. Hindmarsh *et al.*, "SUNDIALS: suite of nonlinear and differential/algebraic equation solvers," *ACM Transactions on Mathematical Software*, vol. 31, no. 3, pp. 363–396, 2005, doi: 10.1145/1089014.1089020.

[7] H. Akaike, "A new look at the statistical model identification," *IEEE Transactions on Automatic Control*, vol. 19, no. 6, pp. 716–723, 1974, doi: 10.1109/TAC.1974.1100705.

[8] G. Schwarz, "Estimating the dimension of a model," *The Annals of Statistics*, vol. 6, no. 2, pp. 461–464, 1978, doi: 10.1214/aos/1176344136.

[9] J. M. Bland and D. G. Altman, "Statistical methods for assessing agreement between two methods of clinical measurement," *The Lancet*, vol. 327, no. 8476, pp. 307–310, 1986, doi: 10.1016/S0140-6736(86)90837-8.

[10] A. Raue *et al.*, "Structural and practical identifiability analysis of partially observed dynamical models by exploiting the profile likelihood," *Bioinformatics*, vol. 25, no. 15, pp. 1923–1929, 2009, doi: 10.1093/bioinformatics/btp358.

[11] C. Kreutz, A. Raue, D. Kaschek, and J. Timmer, "Profile likelihood in systems biology," *The FEBS Journal*, vol. 280, no. 11, pp. 2564–2571, 2013, doi: 10.1111/febs.12276.

[12] F.-G. Wieland, A. L. Hauber, M. Rosenblatt, C. Tönsing, and J. Timmer, "On structural and practical identifiability," *Current Opinion in Systems Biology*, vol. 25, pp. 60–69, 2021, doi: 10.1016/j.coisb.2021.03.005.

[13] C. R. Harris *et al.*, "Array programming with NumPy," *Nature*, vol. 585, pp. 357–362, 2020, doi: 10.1038/s41586-020-2649-2. Version 2.5.3.